Alexander Navarre

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Professional Summary

Specialist in Bayesian statistical modeling and high-dimensional data analysis, with expertise in Python and scientific computing. Data scientist with a PhD in astrophysics, with a strong foundation in data analysis, data visualization, and statistical modeling. Proven ability to present complex results to both technical and non-technical audiences.

Skills

- Python / C / C++ / SQL
- Mathematical Modeling / Statistics
- Bayesian Inference / MCMC
- Windows / Linux
- Machine Learning / Scikit-Learn
- Experimental Design / Hypothesis Testing
- Jupyter / Pandas / Numpy
- Git / Github
- Microsoft Azure / Cloud Resources
- Tableau / Matplotlib / Seaborn
- Scientific Writing / Reporting
- LLM Evaluation & Benchmarking

Work History

Inventory and Product Specialist, 01/2026 to Present
Ben's Cycles – Milwaukee, WI

- Researched, priced, and listed cycling inventory across modern and vintage product lines using market data and condition assessment.
- Managed end-to-end product pipeline for an expanded catalog following acquisition of Euro-Asia Imports.
- Performed product photography and post-processing image editing for e-commerce listings.
- Applied systematic research and structured data workflows to support accurate, competitive pricing decisions.

AI Training Specialist, 05/2025 to 01/2026
Handshake AI – Remote, USA

- Designed high-difficulty prompts in physics and astronomy to test the limits of large AI models via text and image-based reasoning.
- Collaborated with other domain experts to ensure prompt clarity and scientific rigor.
- Provided structured feedback on peer-submitted prompts to enhance quality, diversity, and consistency in relevant failure modes.
- Designed coding evaluation tasks with defined inputs and expected outputs to benchmark LLM code generation capabilities across problem types.

Postdoctoral Researcher, 05/2024 to 08/2024
University of Cincinnati – Cincinnati, OH

- Built and validated a Python modeling pipeline for morphological analysis of gravitationally lensed galaxies, tested across 5 distinct observational fields.
- Identified and documented critical limitations of code for specific data quality.
- Created installation and user guides for research software programs for current and future collaboration members.
- Mentored junior researchers, enhancing their skills and supporting their career development.

Graduate Research Assistant, 10/2019 to 05/2024
University of Cincinnati – Cincinnati, OH

- Applied MCMC-based Bayesian modeling and linear algebraic models to high-dimensional spectral and imaging data from HST and JWST to constrain physical parameters of high-redshift galaxies.
- Authored over 6000 lines of custom Python code for spectral and image analysis.
- Published a first-author research paper in The Astrophysical Journal.
- Presented research findings at conferences and workshops, fostering professional growth and collaboration within the academic community.
- Contributed to the SGAS collaboration, a multi-institutional research team, by performing specialized spectral and morphological analysis and supporting shared data pipelines.

Education

Ph.D.: Physics, 05/2024
University of Cincinnati - Cincinnati, OH

- Subfield: Astrophysics
- 3.8 GPA
- Professional Development: 6th Indo-French Astronomy School: Treasures in the Voxels
- Dissertation: Clumpy vs. Extended Lyman Alpha Emitters at High Redshift

Bachelor of Science: Physics, 05/2018
University of Illinois At Urbana-Champaign - Champaign, IL

- Minor in Astronomy
- 3.4 GPA

CERTIFICATIONS

Data Science Professional Certificate, IBM Data Science via Coursera, 2024-01-01 Certificate ID: FL25WZXTQ58N

Publications

Resolving Clumpy vs. Extended Lyα In Strongly-Lensed, High-Redshift Lyα Emitters
Published in: The Astrophysical Journal
Alexander Navarre et al 2024 ApJ **962** 175